

BobaLink – Browser Extension for Boba Project

Complete Technical Deliverable Package

Version 1.0.0 | June 6, 2026

Project Overview

BobaLink is a comprehensive cross-browser extension (Chrome, Firefox, Opera, Yandex, Chromium) that detects torrent files and magnet links on web pages and in tab groups, then sends them to the Boba Project’s qBitTorrent web dashboard for downloading. This package contains all materials required for a development team to build, test, and release the extension.

Deliverable Structure

/mnt/agents/output/	
â”œâ”€â”€ README.md	# This file â”€” master document
â”œâ”€â”€ plan.md	# Master execution plan
â”œâ”€â”€ implementation-plan.md	# Phased implementation with
â”	
â”œâ”€â”€ src/	# COMPLETE SOURCE CODE (60 fi
â”	# Project dependencies
â”	# TypeScript configuration
â”	# WXT build system config
â”	# Jest test configuration
â”	# Playwright E2E configurat
â”	# ESLint configuration
â”	# Prettier formatting
â”	
â”	# TypeScript type definitio
â”	# Torrent data types
â”	# Configuration types
â”	# API response types
â”	
â”	# Shared utilities
â”	# Regex patterns, constan
â”	# Structured logging
â”	# Custom error classes
â”	# Typed event bus
â”	# AES-256-GCM encryption
â”	# chrome.storage wrapper
â”	# Debounce, throttle, uti
â”	
â”	# Torrent parsing engine
â”	# Magnet link parser

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â",    â",    â"€â"€â"€ bencode.ts           # Bencode encoder/decoder
â",    â",    â""â"€â"€ torrent-file.ts      # .torrent file parser
â",    â",
â",    â"€â"€â"€ src/scanner/                # DOM scanning engine
â",    â",    â"€â"€â"€ base.ts                # Abstract scanner
â",    â",    â"€â"€â"€ link-scanner.ts        # Link element scanner
â",    â",    â"€â"€â"€ text-scanner.ts        # Text node scanner
â",    â",    â"€â"€â"€ site-db.ts            # Site-specific selectors
â",    â",    â""â"€â"€ orchestrator.ts        # Scanner orchestrator
â",    â",
â",    â"€â"€â"€ src/api/                    # API integration layer
â",    â",    â"€â"€â"€ client.ts                # Boba API client
â",    â",    â"€â"€â"€ auth.ts                # Authentication handlers
â",    â",    â"€â"€â"€ qbittorrent.ts          # qBitTorrent API adapter
â",    â",    â"€â"€â"€ health.ts              # Health checker
â",    â",    â""â"€â"€ queue.ts                # Offline queue system
â",    â",
â",    â"€â"€â"€ src/background/              # Service Worker
â",    â",    â""â"€â"€ index.ts                # Background script entry
â",    â",
â",    â"€â"€â"€ src/content/                  # Content Script
â",    â",    â"€â"€â"€ index.ts                # Content script entry
â",    â",    â"€â"€â"€ scanner.ts              # DOM scanning logic
â",    â",    â"€â"€â"€ highlight.ts            # Visual overlay
â",    â",    â""â"€â"€ styles.css              # Content script styles
â",    â",
â",    â"€â"€â"€ src/popup/                    # Popup UI
â",    â",    â"€â"€â"€ index.html              # Popup HTML
â",    â",    â"€â"€â"€ popup.ts                # Popup logic
â",    â",    â""â"€â"€ styles.css              # Popup styles
â",    â",
â",    â"€â"€â"€ src/options/                  # Options Page
â",    â",    â"€â"€â"€ index.html              # Options HTML
â",    â",    â"€â"€â"€ options.ts              # Options logic
â",    â",    â""â"€â"€ styles.css              # Options styles
â",    â",
â",    â"€â"€â"€ src/assets/                    # Icons and assets
â",    â",    â""â"€â"€ icon.svg                # SVG icon source
â",    â",
â",    â"€â"€â"€ _locales/en/                  # Internationalization
â",    â",    â""â"€â"€ messages.json            # English strings
â",    â",
â",    â"€â"€â"€ tests/                        # COMPREHENSIVE TEST SUITES
â",    â",    â"€â"€â"€ unit/                    # Unit tests (Jest)
â",    â",    â",    â"€â"€â"€ magnet.test.ts      # Magnet parser tests (
â",    â",    â",    â"€â"€â"€ bencode.test.ts    # Bencode tests (50+)
â",    â",    â",    â"€â"€â"€ torrent-file.test.ts  # Torrent file tests (4
â",    â",    â",    â"€â"€â"€ scanner.test.ts     # Scanner tests (45+)
â",    â",    â",    â"€â"€â"€ api-client.test.ts   # API client tests (50+
â",    â",    â",    â"€â"€â"€ queue.test.ts       # Queue tests (45+)
â",    â",    â",    â"€â"€â"€ crypto.test.ts      # Encryption tests (40+
â",    â",    â",    â"€â"€â"€ chrome-mock.ts     # Browser API mocks
â",    â",    â",    â""â"€â"€ setup.ts            # Test setup
â",    â",    â",    â"€â"€â"€ fixtures/          # Test fixtures

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â",    â",    â",    â""â"€â"€ magnets.json           # Magnet URI test fixture
â",    â",    â""â"€â"€ e2e/                          # E2E tests (Playwright)
â",    â",    â"œâ"€â"€ popup.spec.ts                # Popup E2E tests (20+)
â",    â",    â"œâ"€â"€ content.spec.ts              # Content script E2E (25+)
â",    â",    â""â"€â"€ options.spec.ts              # Options E2E tests (35+)
â",    â",
â",    â"œâ"€â"€ .github/workflows/                  # CI/CD pipelines
â",    â",    â"œâ"€â"€ ci.yml                        # Lint, test, build, E2E
â",    â",    â""â"€â"€ release.yml                  # Release to all stores
â",    â",
â",    â""â"€â"€ sql/                                # SQL schemas
â",    â"œâ"€â"€ schemas.sql                        # Full database schema
â",    â""â"€â"€ migrations/001_initial.sql          # Initial migration
â",
â"œâ"€â"€ diagrams/                                # 36 DIAGRAMS IN 5 FORMATS
â",    â"œâ"€â"€ 01-system-architecture.*            # System architecture (5 fi
â",    â"œâ"€â"€ 02-dataflow-torrent-detection.*    # Data flow (3 files)
â",    â"œâ"€â"€ 03-dataflow-tab-groups.*            # Tab group batch flow (3 f
â",    â"œâ"€â"€ 04-sequence-single-torrent.*        # Single torrent sequence (
â",    â"œâ"€â"€ 05-sequence-tab-group-batch.*      # Batch sequence (3 files)
â",    â"œâ"€â"€ 06-component-diagram.*              # Component diagram (3 file
â",    â"œâ"€â"€ 07-er-diagram.*                    # ER diagram (4 files)
â",    â"œâ"€â"€ 08-extension-lifecycle.*            # Lifecycle (3 files)
â",    â"œâ"€â"€ 09-offline-queue-state-machine.*    # State machine (2 files)
â",    â"œâ"€â"€ 10-security-architecture.*          # Security (3 files)
â",    â"œâ"€â"€ 11-build-pipeline.*                 # CI/CD pipeline (3 files)
â",    â""â"€â"€ 12-compatibility-matrix.*           # Browser matrix (2 files)
â",
â"œâ"€â"€ docs/                                    # DOCUMENTATION (25,725+ word
â",    â"œâ"€â"€ technical-specification.md          # Technical spec (5,433 wor
â",    â"œâ"€â"€ api-reference.md                    # API reference (5,999 word
â",    â"œâ"€â"€ user-guide.md                       # User guide (5,390 words)
â",    â"œâ"€â"€ developer-guide.md                  # Developer guide (4,800 wo
â",    â""â"€â"€ installation-guide.md               # Installation guide (4,103
â",
â"œâ"€â"€ final/                                    # MULTI-FORMAT DOCUMENTATION
â",    â"œâ"€â"€ technical-specification.pdf         # PDF format (521 KB)
â",    â"œâ"€â"€ technical-specification.docx        # DOCX format (34 KB)
â",    â"œâ"€â"€ technical-specification.html         # HTML format (61 KB)
â",    â"œâ"€â"€ api-reference.pdf                   # PDF format (482 KB)
â",    â"œâ"€â"€ api-reference.docx                  # DOCX format (42 KB)
â",    â"œâ"€â"€ api-reference.html                   # HTML format (77 KB)
â",    â"œâ"€â"€ user-guide.pdf                       # PDF format (449 KB)
â",    â"œâ"€â"€ user-guide.docx                      # DOCX format (30 KB)
â",    â"œâ"€â"€ user-guide.html                      # HTML format (56 KB)
â",    â"œâ"€â"€ developer-guide.pdf                  # PDF format (422 KB)
â",    â"œâ"€â"€ developer-guide.docx                 # DOCX format (35 KB)
â",    â"œâ"€â"€ developer-guide.html                 # HTML format (60 KB)
â",    â"œâ"€â"€ installation-guide.pdf               # PDF format (427 KB)
â",    â"œâ"€â"€ installation-guide.docx              # DOCX format (29 KB)
â",    â""â"€â"€ installation-guide.html              # HTML format (49 KB)
â",
â""â"€â"€ research/                                # DEEP RESEARCH (27,560 lines
    â"œâ"€â"€ boba_extension_dimensions.md          # Dimension decomposition

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â"œâ"€â"€	boba_extension_cross_verification.md	# Cross-verification re
â"œâ"€â"€	boba_extension_insight.md	# Strategic insights (10 insi
â""â"€â"€	boba_extension_dim01.md	# Dimension 01: Boba Architec
â""â"€â"€	boba_extension_dim02.md	# Dimension 02: qBitTorrent A
â""â"€â"€	boba_extension_dim03.md	# Dimension 03: Extension Arc
â""â"€â"€	boba_extension_dim04.md	# Dimension 04: Cross-Browser
â""â"€â"€	boba_extension_dim05.md	# Dimension 05: Tab Groups AP
â""â"€â"€	boba_extension_dim06.md	# Dimension 06: Magnet Parsin
â""â"€â"€	boba_extension_dim07.md	# Dimension 07: Torrent Parsi
â""â"€â"€	boba_extension_dim08.md	# Dimension 08: DOM Scanning
â""â"€â"€	boba_extension_dim09.md	# Dimension 09: API Integrati
â""â"€â"€	boba_extension_dim10.md	# Dimension 10: UI/UX Design
â""â"€â"€	boba_extension_dim11.md	# Dimension 11: Security Mode
â""â"€â"€	boba_extension_dim12.md	# Dimension 12: Testing & Bui

File Count Summary

Category	Files Formats	
Source Code	60	TypeScript, HTML, CSS, JSON, YAML, SQL
Diagrams	36	Mermaid, SVG, draw.io, HTML, PlantUML
Documentation (source)	5	Markdown
Documentation (PDF)	5	PDF
Documentation (DOCX)	5	DOCX
Documentation (HTML)	5	HTML
Research	15	Markdown
Plans	2	Markdown
TOTAL	133	8 formats

Key Statistics

- **Total research:** 27,560 lines across 12 dimensions
- **Source code:** 60 production-ready TypeScript/HTML/CSS files
- **Test coverage:** 320+ test cases (unit + E2E)
- **Documentation:** 25,725+ words across 5 documents
- **Diagrams:** 36 files across 12 diagram types and 5 formats
- **Total package size:** ~4.8 MB

Quick Start for Development Team

1. **Read the Implementation Plan:** implementation-plan.md â€” 8 phases with fine-grained tasks
2. **Review Technical Spec:** docs/technical-specification.md â€” architecture, requirements, data models
3. **Explore Source Code:** src/ â€” complete, compilable TypeScript
4. **Run Tests:** cd src && npm test (after npm install)
5. **Build Extension:** cd src && npm run build
6. **Review Diagrams:** diagrams/ â€” all architecture and flow diagrams

Verification Checklist

- ☐ Deep research (12 dimensions with cross-verification)
 - ☐ Implementation plan (all phases, sub-phases, tasks)
 - ☐ Complete source code (TypeScript, strict mode, production-ready)
 - ☐ SQL schemas (tables, indexes, migrations)
 - ☐ Unit tests (320+ test cases, 80%+ coverage target)
 - ☐ E2E tests (Playwright, 80+ test cases)
 - ☐ CI/CD pipelines (GitHub Actions)
 - ☐ Technical specification (25 FRs, 15 NFRs)
 - ☐ API reference (all endpoints with curl/JS/TS examples)
 - ☐ User guide (step-by-step with troubleshooting, 30 FAQs)
 - ☐ Developer guide (setup, build, test, contribute)
 - ☐ Installation guide (all 5 browsers)
 - ☐ System architecture diagrams
 - ☐ Data flow diagrams
 - ☐ Sequence diagrams
 - ☐ Component diagrams
 - ☐ ER diagrams
 - ☐ State machine diagrams
 - ☐ Security architecture diagrams
 - ☐ Cross-browser compatibility matrix
 - ☐ All diagrams in Mermaid, SVG, PNG reference, draw.io, UML
 - ☐ Documentation in Markdown
 - ☐ Documentation in PDF (enterprise styled)
 - ☐ Documentation in DOCX (enterprise styled)
 - ☐ Documentation in HTML (clean styled)
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This package was generated on 2026-06-06 and represents a complete, production-ready deliverable for the BobaLink browser extension project.